Example 2 *The following table shows the lives in hours of four brands of electric lamps:*

Brand

A : 1610, 1610, 1650, 1680, 1700, 1720, 1800
B : 1580, 1640, 1640, 1700, 1750
C : 1460, 1550, 1600, 1620, 1640, 1660, 1740, 1820
D : 1510, 1520, 1530, 1570, 1600, 1680

Perform an analysis of variance and test the homogeneity of the mean lives of the four brands of lamps.

Soln:- Here mean of all values is 1640.
So we subtract 1640 from all the values
and we perform the ANOVA Table.

Brands	lives (x_{ij})								T_i	n_i	T_i^2/n_i
A	−30	−30	10	40	60	80	160	−	290	7	12014
B	−60	0	0	60	110	−	−	−	110	5	2420
C	−180	−90	−40	−20	0	20	100	180	−30	8	113
D	−130	−120	−110	−70	−40	40	−	−	−430	6	30817.
									T = −60	N = 26	45364

$$\sum\sum x_{ij}^2 = \left((-30)^2 + (-30)^2 + 10^2 + 40^2 + 60^2 + 80^2 + 160^2\right)$$
$$+\left((-60)^2 + 0 + 0 + 60^2 + 110^2\right)$$
$$+\left((-180)^2 + (-90)^2 + (-40)^2 + (-20)^2 + 0 + 20^2 + 100^2 + 180^2\right)$$
$$+\left((-130)^2 + (-120)^2 + (-110)^2 + (-70)^2 + (-40)^2 + 40^2\right)$$
$$= 1{,}95{,}200.$$

$$Q = \sum\sum x_{ij}^2 - \frac{T^2}{N} = 195200 - \frac{(-60)^2}{26} = 1{,}95{,}061.5$$

$$Q_1 = \sum \frac{T_i^2}{n_i} - \frac{T^2}{N} = 45364 - \frac{3600}{26} = 45\,225.5$$

$$Q_2 = Q - Q_1 = 195061.5 - 45225.5 = 1{,}49{,}836$$

ANOVA TABLE

S.V	S.S	d.f	M.S	F values
Between classes	$Q_1 = 45225.5$	$h-1 = 4-1 = 3$	$\frac{Q_1}{h-1} = 15075.17$	$\frac{15075.17}{6810.72} = 2.21$
within classes	$Q_2 = 149836$	$N-h = 26-4 = 22$	$\frac{Q_2}{N-h} = 6810.72$	

F_{value} follows F-distribution with d.f $(h-1, N-h)$
$= (3, 22)$

$F_{0.05} = 3.06.$

H_0: The means of lives of the four brands are homogeneous.

Since $F_{value} < F_{0.05}$ Accept H_0.

ANOVA for 2 factor classification

$S.V.$	$S.S.$	$d.f.$	$M.S.$	F
Between rows	Q_1	$h-1$	$Q_1/(h-1)$	$\left[\dfrac{Q_1/(h-1)}{Q_3/(h-1)(k-1)}\right]^{\pm 1}$
Between columns	Q_2	$k-1$	$Q_2/(k-1)$	$\left[\dfrac{Q_2/(k-1)}{Q_3/(h-1)(k-1)}\right]^{\pm 1}$
Residual	Q_3	$(h-1)(k-1)$	$Q_3/(h-1)(k-1)$	—
Total	Q	$hk-1$	—	—

Note *The following working formulas that can be easily derived may be used to compute Q, Q_1, Q_2 and Q_3:*

1. $Q = \sum \sum x_{ij}^2 - \dfrac{T^2}{N}$, where $T = \sum \sum x_{ij}$
2. $Q_1 = \dfrac{1}{k} \sum T_i^2 - \dfrac{T^2}{N}$, where $T_i = \sum_{j=1}^{k} x_{ij}$
3. $Q_2 = \dfrac{1}{h} \sum T_j^2 - \dfrac{T^2}{N}$, where $T_j = \sum_{i=1}^{h} x_{ij}$
4. $Q_3 = Q - Q_1 - Q_2$

It may be verified that $\sum_i T_i = \sum_j T_j = T.$

Example 5 *Three varieties of a crop are tested in a randomised block design with four replications, the layout being as given below: The yields are given in kilograms. Analyse for significance*

C48	A51	B52	A49
A47	B49	C52	C51
B49	C53	A49	B50

Soln:- Re writing the given data yields the following table.

Blocks	Crops A	B	C
1	47	49	48
2	51	49	53
3	49	52	52
4	49	50	51

We subtract 50 from each of the values.

Blocks	A	B	C	T_i	k_i	T_i^2/k_i	$\sum_j x_{ij}^2$
1	-3	-1	-2	-6	3	12	14
2	1	-1	3	3	3	3	11
3	-1	2	2	3	3	3	9
4	-1	0	1	0	3	0	2
T_j	-4	0	4	$T=0$		$\sum \frac{T_i^2}{k_i} = 18$	36
T_j^2/h	$16/4=4$	$0/4=0$	$16/4=4$	$\sum \frac{T_j^2}{h} = 8$			
$\sum_i x_{ij}^2$	12	6	18	36			

$$Q = \sum\sum x_{ij}^2 - \frac{T^2}{N} = 36 - \frac{0^2}{N} = 36$$

$$Q_1 = \frac{\sum T_i^2}{k} - \frac{T^2}{N} = 18 - 0 = 18$$

$$Q_2 = \frac{\sum T_j^2}{h} - \frac{T^2}{N} = 8 - 0 = 8$$

$$Q_3 = Q - Q_1 - Q_2 = 36 - 18 - 8 = 10$$

ANOVA Table

S V	S.S	d.f	M.S	F value
Between rows	$Q_1 = 18$	$h-1 = 4-1 = 3$	$\frac{Q_1}{h-1} = 6$	$\frac{6}{1.67} = 3.6$
Between columns	$Q_2 = 8$	$k-1 = 3-1 = 2$	$\frac{Q_2}{k-1} = 4$	$\frac{4}{1.67} = 2.4$
Residual	$Q_3 = 10$	$(h-1)(k-1) = 6$	$\frac{Q_3}{(h-1)(k-1)} = 1.67$	

From F table $F_{0.05,(3,6)} = 4.76$

$F_{0.05,(2,6)} = 5.14$

H_a: There is no significant difference between rows

H_b: There is no significant difference between columns.

Since Fvalue $= 3.6 < 4.76$ accept H_a.

Fvalue $= 2.4 < 5.14$ accept H_b.

The varieties of the crop do not differ significantly with respect to the yield.

Example 6 *Five breeds of cattle B_1, B_2, B_3, B_4, B_5 were fed on four different rations R_1, R_2, R_3, R_4. Gains in weight in kg over a given period were recorded and given below:*

	B_1	B_2	B_3	B_4	B_5
R_1	1.9	2.2	2.6	1.8	2.1
R_2	2.5	1.9	2.3	2.6	2.2
R_3	1.7	1.9	2.2	2.0	2.1
R_4	2.1	1.8	2.5	2.3	2.4

Is there a significant difference between (i) breeds and (ii) rations?

ANOVA for 3 factor classification

S.V.	*S.S.*	*d.f.*	*M.S.*	*F*
Between rows	Q_1	$n-1$	$Q_1/(n-1)=M_1$	$\left(\dfrac{M_1}{M_4}\right)^{\pm 1}$
Between columns	Q_2	$n-1$	$Q_2/(n-1)=M_2$	$\left(\dfrac{M_2}{M_4}\right)^{\pm 1}$
Between letters	Q_3	$n-1$	$Q_3/(n-1)=M_3$	$\left(\dfrac{M_3}{M_4}\right)^{\pm 1}$
Residual	Q_4	$(n-1)(n-2)$	$Q_4/(n-1)(n-2)=M_4$	—
Total	Q	n^2-1	—	—

Note *The following working formulas may be used to compute the Q's:*

1. $Q = \sum \sum x_{ij}^2 - \frac{T^2}{n^2}$, *where* $T = \sum \sum x_{ij}$

2. $Q_1 = \frac{1}{n} \sum T_i^2 - \frac{T^2}{n^2}$, *where* $T_i = \sum_{j=1}^{n} x_{ij}$

3. $Q_2 = \frac{1}{n} \sum T_j^2 - \frac{T^2}{n^2}$, *where* $T_j = \sum_{i=1}^{n} x_{ij}$

4. $Q_3 = \frac{1}{n} \sum T_k^2 - \frac{T^2}{n^2}$, *where* T_k *is the sum of all* x_{ij}*'s receiving the* k^{th} *treatment.*

5. $Q_4 = Q - Q_1 - Q_2 - Q_3$.

Also $T = \sum_i T_i = \sum_j T_j = \sum_k T_k$

Example 9 *The following data resulted from an experiment to compare three burners B_1, B_2 and B_3. A Latin square design was used as the tests were made on 3 engines and were spread over 3 days.*

	Engine 1	*Engine 2*	*Engine 3*
Day 1	$B_1 - 16$	$B_2 - 17$	$B_3 - 20$
Day 2	$B_2 - 16$	$B_3 - 21$	$B_1 - 15$
Day 3	$B_3 - 15$	$B_1 - 12$	$B_2 - 13$

Test the hypothesis that there is no difference between the burners.

Soln:- we subtract 16 from each of the given Values.

	E_1	E_2	E_3	T_i	$\frac{T_i^2}{n}$	$\sum_j x_{ij}^2$
D_1	0	1	4	5	$25/3 = 8.33$	17
D_2	0	5	-1	4	$16/3 = 5.33$	26
D_3	-1	-4	-3	-8	$64/3 = 21.33$	26
				$\sum T_i = 1$	$\sum \frac{T_i^2}{n} = 35$	$\sum x_{ij}^2 = 69$
T_j	-1	2	0			
T_j^2/n	$1/3 = 0.33$	$4/3 = 1.33$	0	$\sum \frac{T_j^2}{n} = 1.66$		
$\sum_i x_{ij}^2$	1	42	26	69.		

Next table is for burners,

Burner	x_k			T_k	T_k^2/n
B_1	0	-1	-4	-5	8.33
B_2	1	0	-3	-2	1.33
B_3	4	5	-1	8	21.33
				$T = \sum T_k = 1.$	$\sum \frac{T_k^2}{n} = 31$

$$Q = \sum\sum x_{ij}^2 - \frac{T^2}{N} = 69 - \frac{1}{9} = 68.89$$

$$Q_1 = \frac{\sum T_i^2}{n} - \frac{T^2}{N} = 35 - \frac{1}{9} = 34.89$$

$$Q_2 = \frac{\sum T_j^2}{n} - \frac{T^2}{N} = 1.67 - 1/9 = 1.56$$

$$Q_3 = \frac{\sum T_k^2}{n} - \frac{T^2}{N} = 31 - 1/9 = 30.89$$

$$Q_4 = Q - Q_1 - Q_2 - Q_3 = 1.55.$$

ANOVA TABLE

S.V	S.S	d.f	M.S	F value
Between rows (days)	$Q_1 = 34.89$	$n-1 = 2$	$\frac{Q_1}{n-1} = 17.445$	$\frac{17.445}{0.775} = 22.51$
Between columns (Engines)	$Q_2 = 1.56$	$n-1 = 2$	$\frac{Q_2}{n-1} = 0.780$	$\frac{0.780}{0.775} = 1.01$
Between Burners	$Q_3 = 30.89$	$n-1 = 2$	$\frac{Q_3}{n-1} = 15.445$	$\frac{15.445}{0.775} = 19.93$
Residual	$Q_4 = 1.55$	$(n-1)(n-2) = 2$	$\frac{Q_4}{(n-1)(n-2)} = 0.775$	

H_a: There is no significant difference between days.

H_b: There is no significant difference between Engines.

H_c: There is no significant difference between burners.

From F table $F_{0.05,(2,2)} = 19$.

(i) $F_{value} = 22.51 > F_{0.05}$ Reject H_a.

(ii) $F_{value} = 1.01 < F_{0.05}$ Accept H_b.

(iii) $F_{value} = 19.93 > F_{0.05}$ Reject H_c.

Example 10 *Analyse the variance in the following Latin square of yields (in kgs) of paddy where A, B, C, D denote the different methods of cultivation*

*D*122	*A*121	*C*123	*B*122
*B*124	*C*123	*A*122	*D*125
*A*120	*B*119	*D*120	*C*121
*C*122	*D*123	*B*121	*A*122

Examine whether the different methods of cultivation have given significantly different yields.

1. Completely Randomised Design (C.R.D.)

Let us suppose that we wish to compare 'h' treatments (use of 'h' different manures) and there are n plots available for the experiment.

Let the ith treatment be replicated (repeated) n_i times, so that $n_1 + n_2 + \ldots + n_h = n$.

The plots to which the different treatments are to be given are found by the following randomisation principle. The plots are numbered from 1 to n serially. n identical cards are taken, numbered from 1 to n and shuffled thoroughly. The numbers on the first n_1 cards drawn randomly give the numbers of the plots to which the first treatment is to be given. The numbers on the next n_2 cards drawn at random give the numbers of the plots to which the second treatment is to be given and so on. This design is called a completely randomised design, which is used when the plots are homogeneous or the pattern of heterogeneity of the plots is unknown.

2. *Randomised Block Design (R.B.D.)*

Let us consider an agricultural experiment using which we wish to test the effect of 'k' fertilizing treatments on the yield of a crop. We assume that we know some information about the soil fertility of the plots. Then we divide the plots into 'h' blocks, according to the soil fertility, each block containing 'k' plots. Thus the plots in each block will be of homogeneous fertility as far as possible.

Within each block, the 'k' treatments are given to the 'k' plots in a perfectly random manner, such that each treatment occurs only once in any block. But the same 'k' treatments are repeated from block to block. This design is called Randomised Block Design.

3. Latin Square Design (L.S.D.)

We consider an agricultural experiment, in which n^2 plots are taken and arranged in the form of an $n \times n$ square, such that the plots in each row will be homogeneous as far as possible with respect to one factor of classification, say, soil fertility and plots in each column will be homogeneous as far as possible with respect to another factor of classification, say, seed quality.

Then n treatments are given to these plots such that each treatment occurs only once in each row and only once in each column. The various possible arrangements obtained in this manner are known as Latin squares of order n. This design of experiment is called the Latin Square Design.

Comparison of RBD and LSD

1. The number of replications of each treatment is equal to the number of treatments in LSD, whereas there is no such restrictions on treatments and replication in RBD.
2. LSD can be performed on a square field, while RBD can be performed either on a square field or a rectangular field.
3. LSD is known to be suitable for the case when the number of treatments is between 5 and 12, whereas RBD can be used for any number of treatments.
4. The main advantage of LSD is that it controls the effect of two extraneous variables, whereas RBD controls the effect of only one extraneous variable. Hence the experimental error is reduced to a larger extent in LSD than in RBD.